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Electronic device for supporting connection with external device and operating method thereof

Abstract

According to an example embodiment, an electronic device includes: a universal serial bus (USB) power delivery integrated circuit (PDIC) configured to determine whether a first pin of a USB port of the electronic device is connected to a power delivery (PD) source; a charging circuitry configured to charge a battery of the electronic device with power supplied from the PD source through a second pin of the USB port; a memory configured to store computer-executable instructions; and a processor configured to execute the instructions by accessing the memory. When the USB PDIC determines that the first pin and the PD source are connected, the instructions may cause the processor to control a voltage of the second pin to be less than a threshold voltage while maintaining communication between the USB PDIC and the PD source through the first pin based on a state of the electronic device. In addition, various example embodiments may be possible.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/KR2022/005145 designating the United States, filed on Apr. 8, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2021-0070032, filed on May 31, 2021, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

1. Field

(1) The instant disclosure generally relates to an electronic device that supports connection with an external electronic device and a method of operating the electronic device, and more particularly, to an electronic device that receives power supplied through a universal serial bus (USB) port and a method of operating the electronic device.

2. Description of Related Art

(2) Various electronic devices such as smartphones, tablet personal computers (PCs), portable multimedia players (PMPs), personal digital assistants (PDAs), laptop PCs, and wearable devices have become popular. These various electronic devices typically have ports that are used for connecting with external electronic devices via wires. The ports are specified according to various standards. Among such various standards, the universal serial bus (USB) standard is the most widely used.

(3) USB is an input/output standard used to connect an electronic device and an external electronic device. Recently, with the introduction of USB Type-C power delivery (PD) technologies, one electronic device may receive power from a USB-connected PD source.

SUMMARY

(4) When an electronic device is connected to a power delivery (PD) source via a universal serial bus (UBS) Type-C port, the configuration channel (CC) line defined in the USB Type-C standard may be physically connected to a pull-up resistor Rp and a pull-down resistor Rd, and virtual bus (VBUS) voltage, which is the output voltage of the PD source, may become 5 volts (V) or greater, so that the electronic device and the PD source may communicate through the CC line.

(5) When the VBUS voltage of the PD source becomes 5 V, voltage required by the electronic device may be output through the CC line, and 20 V may be output.

(6) In existing USB Type-C, CC communication may be possible only when the minimum VBUS voltage of 5V is maintained. In other words, the VBUS voltage of 5 V may need to be maintained at all times even though the VBUS voltage is not required, and accordingly standby power may continue to be consumed.

(7) According to an example embodiment, an electronic device includes: a USB power delivery integrated circuit (PDIC) configured to determine whether a first pin of a USB port of the electronic device is connected to a PD source; a charging circuitry configured to charge a battery of the electronic device with power supplied from the PD source through a second pin of the USB port; a memory configured to store computer-executable instructions; and a processor configured to execute the instructions by accessing the memory, wherein the instructions cause the USB PDIC to control a voltage of the second pin to be less than a threshold voltage while maintaining communication through the first pin with the PD source based on a state of the electronic device when the USB PDIC determines that the first pin and the PD source are connected.

(8) According to an example embodiment, a method of operating an electronic device includes: determining whether a first pin of a USB port of the electronic device is connected to a PD source; and controlling a voltage of a second pin of the USB port to be less than a threshold voltage while maintaining communication through the first pin with the PD source based on a state of the electronic device, when it is determined that the first pin and the PD source are connected, wherein a battery of the electronic device is configured to receive power supplied from the PD source through the second pin.

(9) According to an example embodiment, a non-transitory computer-readable storage medium may store instructions that, when executed by an electronic device, cause the electronic device to:

determine whether a first pin of a USB port of the electronic device is connected to a PD source; and control a voltage of a second pin of the USB port to be less than a threshold voltage while maintaining communication through the first pin with the PD source based on a state of the electronic device, when it is determined that the first pin and the power supply device are connected, wherein a battery of the electronic device is configured to receive power supplied from the PD source through the second pin.

(10) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. **1** is a block diagram illustrating an electronic device in a network environment according to an example embodiment;

(3) FIG. **2** is a block diagram of a power management module and a battery according to an example embodiment;

(4) FIG. **3** is a block diagram illustrating an electronic device according to an example embodiment;

(5) FIG. **4** is a diagram illustrating pins constituting a universal serial bus (USB) port according to an example embodiment;

(6) FIGS. **5**A and **5**B are diagrams illustrating an operation in which an electronic device receives power, according to an example embodiment;

(7) FIG. **6** is a flowchart illustrating an operation of an electronic device according to an example embodiment; and

(8) FIG. **7** is a flowchart illustrating an operation of controlling a virtual bus (VBUS) voltage based on a state of an electronic device according to an example embodiment.

DETAILED DESCRIPTION

(9) Hereinafter, certain example embodiments will be described in greater detail with reference to the accompanying drawings. When describing the example embodiments with reference to the accompanying drawings, like reference numerals refer to like elements and a repeated description related thereto will be omitted.

(10) According to certain example embodiments, an electronic device that monitors a state of the electronic device to control VBUS to be 0 V when power supply from a PD source is not required may be provided, so as to minimize standby power through CC communication control by controlling VBUS to be less than 5 V.

(11) According to certain example embodiments, an electronic device that controls VBUS voltage of a PD source to 0 V based on a state of the electronic device may be provided. According to certain example embodiments, an electronic device that controls standby power through CC communication when VBUS voltage of a PD source is less than a threshold voltage may be provided.

(12) <Electronic Device>

(13) FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to an example embodiment.

(14) Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or communicate with at least one of an electronic device **104** or a server

108 via a second network **199** (e.g., a long-range wireless communication network). According to an example embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an example embodiment, the electronic device **101** may include a processor **120**, a memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, and a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some example embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some example embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be integrated as a single component (e.g., the display module **160**).

(15) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** connected to the processor **120**, and may perform various data processing or computation. According to an example embodiment, as at least a part of data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. According to an example embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)) or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently of, or in conjunction with the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121** or to be specific to a specified function. The auxiliary processor **123** may be implemented separately from the main processor **121** or as a part of the main processor **121**.

(16) The auxiliary processor **123** may control at least some of functions or states related to at least one (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) of the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state or along with the main processor **121** while the main processor **121** is an active state (e.g., executing an application). According to an example embodiment, the auxiliary processor **123** (e.g., an ISP or a CP) may be implemented as a portion of another component (e.g., the camera module **180** or the communication module **190**) that is functionally related to the auxiliary processor **123**. According to an example embodiment, the auxiliary processor **123** (e.g., an NPU) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed by, for example, the electronic device **101** in which artificial intelligence is performed, or performed via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, for example, supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The AI model may include a plurality of artificial neural network layers. An artificial neural network may include, for example, a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), and a bidirectional recurrent deep neural network (BRDNN), a deep Q-network, or a combination of two or more thereof, but is not limited thereto. The AI model may additionally or alternatively include a software structure other than the hardware structure.

(17) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for

example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.
(18) The program **140** may be stored as software in the memory **130**, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
(19) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
(20) The sound output module **155** may output a sound signal to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used to receive an incoming call. According to an example embodiment, the receiver may be implemented separately from the speaker or as a part of the speaker.
(21) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a control circuit for controlling a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, the hologram device, and the projector. According to an example embodiment, the display module **160** may include a touch sensor adapted to sense a touch, or a pressure sensor adapted to measure an intensity of a force incurred by the touch.
(22) The audio module **170** may convert a sound into an electric signal or vice versa. According to an example embodiment, the audio module **170** may obtain the sound via the input module **150** or output the sound via the sound output module **155** or an external electronic device (e.g., an electronic device **102** such as a speaker or a headphone) directly or wirelessly connected to the electronic device **101**.
(23) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and generate an electric signal or data value corresponding to the detected state. According to an example embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, a Hall sensor, or an illuminance sensor.
(24) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an example embodiment, the interface **177** may include, for example, a high-definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
(25) The connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected to an external electronic device (e.g., the electronic device **102**). According to an example embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).
(26) The haptic module **179** may convert an electric signal into a mechanical stimulus (e.g., a vibration or a movement) or an electrical stimulus which may be recognized by a user via his or her tactile sensation or kinesthetic sensation. According to an example embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.
(27) The camera module **180** may capture a still image and moving images. According to an example embodiment, the camera module **180** may include one or more lenses, image sensors, ISPs, or flashes.
(28) The power management module **188** may manage power supplied to the electronic device **101**. According to an example embodiment, the power management module **188** may be implemented

as, for example, at least a part of a power management integrated circuit (PMIC).
(29) The battery **189** may supply power to at least one component of the electronic device **101**. According to an example embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.
(30) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently of the processor **120** (e.g., an AP) and that support a direct (e.g., wired) communication or a wireless communication. According to an example embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module, or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., a LAN or a wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the SIM **196**.
(31) The wireless communication module **192** may support a 5G network after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., a mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (MIMO), full dimensional MIMO (FD-MIMO), an array antenna, analog beam-forming, or a large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an example embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.
(32) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an example embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an example embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected by, for example, the communication module **190** from the plurality of antennas. The signal or the power may be transmitted or received between the

communication module **190** and the external electronic device via the at least one selected antenna. According to an example embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as a part of the antenna module **197**.

(33) According to certain example embodiments, the antenna module **197** may form a mmWave antenna module. According to an example embodiment, the mmWave antenna module may include a PCB, an RFIC disposed on a first surface (e.g., a bottom surface) of the PCB or adjacent to the first surface and capable of supporting a designated a high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., a top or a side surface) of the PCB, or adjacent to the second surface and capable of transmitting or receiving signals in the designated high-frequency band.

(34) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(35) According to an example embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of the same type as or a different type from the electronic device **101**. According to an example embodiment, all or some of operations to be executed by the electronic device **101** may be executed at one or more external electronic devices (e.g., the external devices **102** and **104**, and the server **108**). For example, if the electronic device **101** needs to perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and may transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an example embodiment, the external electronic device **104** may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an example embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(36) FIG. **2** is a block diagram **200** of a power management module **188** and a battery **189** according to an example embodiment.

(37) Referring to FIG. **2**, the power management module **188** may include charging circuitry **210**, a power adjuster **220**, or a power gauge **230**. The charging circuitry **210** may charge the battery **189** using power supplied from an external power source outside the electronic device **101**. According to an example embodiment, the charging circuitry **210** may select a charging scheme (e.g., normal charging or quick charging) based at least in part on the type of the external power source (e.g., power outlet, USB charger, or wireless charger), magnitude of power suppliable from the external power source (e.g., about 20 Watts or more), or an attribute of the battery **189**, and may charge the battery **189** using the selected charging scheme. The external power source may be connected with the electronic device **101**, for example, directly via the connecting terminal **178** or wirelessly via the antenna module **197**.

(38) The power adjuster **220** may generate a plurality of powers having different voltage levels or different current levels by adjusting the voltage level or the current level of the power supplied from the external power source or the battery **189**. The power adjuster **220** may adjust the voltage level or the current level of the power supplied from the external power source or the battery **189** into different voltage levels or current levels appropriate for each of the various components included in the electronic device **101**. According to an example embodiment, the power adjuster **220** may be implemented in the form of a low drop out (LDO) regulator or a switching regulator. The power gauge **230** may measure use state information about the battery **189** (e.g., capacity, the number of times the battery has been charged or discharged, voltage, or temperature of the battery **189**).
(39) The power management module **188** may determine, using, for example, the charging circuitry **210**, the power adjuster **220**, or the power gauge **230**, charging state information (e.g., lifetime, over voltage, low voltage, over current, over charge, over discharge, overheat, short, or swelling) related to the charging of the battery **189** based at least in part on the measured use state information about the battery **189**. The power management module **188** may determine whether the state of the battery **189** is normal or abnormal based at least in part on the determined charging state information. If the state of the battery **189** is determined to abnormal, the power management module **188** may adjust the charging of the battery **189** (e.g., reduce the charging current or voltage, or stop the charging). According to an example embodiment, at least some of the functions of the power management module **188** may be performed by a control device not shown in FIG. **2** (e.g., the processor **120**).
(40) According to an example embodiment, the battery **189** may include a protection circuit module (PCM) **240**. The PCM **240** may perform one or more of various functions (e.g., pre-cutoff function) to prevent performance deterioration of, or damage to, the battery **189**. The PCM **240**, additionally or alternatively, may be configured as at least part of a battery management system (BMS) capable of performing various functions including cell balancing, measurement of battery capacity, count of the number of charging or discharging cycles, measurement of temperature, or measurement of voltage.
(41) According to an example embodiment, at least part of the charging state information or use state information regarding the battery **189** may be measured using an appropriate sensor (e.g., temperature sensor) of the sensor module **176**, the power gauge **230**, or the power management module **188**. According to an example embodiment, the appropriate sensor (e.g., temperature sensor) of the sensor module **176** may be included as part of the PCM **240**, or may be disposed near the battery **189** as a separate device.
(42) The electronic devices according to certain example embodiments may be various types of electronic devices. The electronic device may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance device. According to an example embodiment of the disclosure, the electronic device is not limited to those described above.
(43) It should be appreciated that certain example embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular example embodiments and include various changes, equivalents, or replacements for a corresponding example embodiment. In connection with the description of the drawings, like reference numerals may be used for similar or related components. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B or C”, “at least one of A, B and C”, and “A, B, or C,” each of which may include any one of the items listed together in the corresponding one of the phrases, or all possible combinations thereof. Terms such as “first”, “second”, or “first” or “second” may simply be used to distinguish the component from other components in question, and may refer to

components in other aspects (e.g., importance or order) is not limited. It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(44) As used in connection with certain example embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an example embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(45) Certain example embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., an internal memory **136** or an external memory **138**) that is readable by a machine (e.g., the electronic device **101** of FIG. **1**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(46) According to an example embodiment, a method according to certain example embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(47) According to certain example embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to certain example embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to certain example embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to certain example embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(48) FIG. **3** is a block diagram illustrating an electronic device according to an example embodiment.

(49) Referring to FIG. **3**, the electronic device **101** may include software **310**, hardware **320**, a USB connector **330**, a memory **130** configured to store computer-executable instructions, and a

processor **120** configured to execute the instructions by accessing the memory **130**. The processor **120** may include a microprocessor or any suitable type of processing circuitry, such as one or more general-purpose processors (e.g., ARM-based processors), a Digital Signal Processor (DSP), a Programmable Logic Device (PLD), an Application-Specific Integrated Circuit (ASIC), a Field-Programmable Gate Array (FPGA), a Graphical Processing Unit (GPU), a video card controller, etc. In addition, it would be recognized that when a general purpose computer accesses code for implementing the processing shown herein, the execution of the code transforms the general purpose computer into a special purpose computer for executing the processing shown herein. Certain of the functions and steps provided in the Figures may be implemented in hardware, software or a combination of both and may be performed in whole or in part within the programmed instructions of a computer. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f), unless the element is expressly recited using the phrase “means for.” In addition, an artisan understands and appreciates that a “processor” or “microprocessor” may be hardware in the claimed disclosure. Under the broadest reasonable interpretation, the appended claims are statutory subject matter in compliance with 35 U.S.C. § 101.
(50) According to an example embodiment, in the memory **130** of the electronic device **101**, a program (e.g., the program **140** of FIG. **1**) to control virtual bus (VBUS) voltage, which is the output voltage of a PD source **350** through a CC pin, to be less than a threshold voltage based on a state of the electronic device **101** may be stored as software. For example, an OS (e.g., the OS **142** of FIG. **1**), middleware (e.g., the middleware **144** of FIG. **1**) or an application **146** may be included in the memory **130**. The instructions stored in the memory **130** may be implemented as one function module in the OS (e.g., the OS **142** of FIG. **1**), implemented in the form of middleware (e.g., the middleware **144** of FIG. **1**), or implemented in the form of a separate application **146**.
(51) According to an example embodiment, the software **310** may execute programs (e.g., the programs **140** of FIG. **1**) such as applications (e.g., the applications **146** of FIG. **1**) for an operation of the electronic device **101**. The software **310** may include a framework **311** and a kernel **312**.
(52) According to an example embodiment, the framework **311** may be a collaborative environment of the software **310** provided such that design and implementation of parts of the software **310** may be reused or shared to facilitate the development of the application **146** of the software **310**. The framework **311** may include a USB framework **311-1** controlling the USB connection that is used to connect with the PD source **350**, and a battery service **311-2** controlling charging and/or discharging of a battery (e.g., the battery **189** of FIG. **1** or the battery **189** of FIG. **2**) included in the electronic device **101**.
(53) According to an example embodiment, the kernel **312** may perform resource allocation for the program **140** running under control of the processor (e.g., processor **120** of FIG. **1**) executing the OS (e.g., the OS **142** of FIG. **1**). The kernel **312** may include a USB driver **312-1** establishing a USB connection, a charging driver **312-2** controlling a charging mode of the battery **189**, and/or a power delivery integrated circuit (PDIC) driver **312-3** that detects resistance applied to the USB connector **330** and is responsible for power delivery (PD) communication when the PD source **350** is connected to the USB connector **330**. The USB driver **312-1** according to an example embodiment may be executed by the processor **120**. The USB driver **312-1** according to an example embodiment may be stored in the memory **130**. That is, the USB driver **312-1** may be stored in the memory **130** and may be executed by the processor **120**.
(54) According to an example embodiment, the hardware **320** may be the physical components of the electronic device **101**. The hardware **320** may include a USB controller **321**, charging circuitry **210** (e.g., the charging circuitry **210** of FIG. **2**), and a USB PDIC **323**.
(55) According to an example embodiment, the USB controller **321** may be included in the processor **120**. The USB controller **321** may perform USB data communication using the D+ line and the D− line included in the USB connector **330**.
(56) According to an example embodiment, the charging circuitry **210** may be included in a PMIC

(e.g., the power management module **188** of FIG. **1**). The charging circuitry **210** may charge the battery **189** with power supplied from the PD source **350**, which is an external power supply device outside the electronic device **101**, via the VBUS line included in the USB connector **330**, and the battery **189** may supply power to at least one component of the electronic device **101**. According to an example embodiment, the charging circuitry **210** may convert power (e.g., 5 V or greater and 20 V or less) supplied from the PD source **350** and may output main power to be supplied to each component of the electronic device **101**. According to an example embodiment, if the electronic device **101** is not connected to the PD source **350**, the charging circuitry **210** may control power to be supplied from the battery **189** to each component of the electronic device **101**.
(57) According to an example embodiment, the USB PDIC **323** may be included in the power management module **188**. The USB PDIC **323** may determine whether the USB PDIC **323** is connected to an external device (e.g., the PD source **350**) and may perform bi-phase marked communication (BMC), through the CC pin included in the USB connector **330**. The BMC may be single wire communication in which signals are transmitted and received using a single wire. According to an example embodiment, the USB PDIC **323** may control the VBUS voltage of the PD source **350** using the CC pin. Specifically, the processor **120** may monitor the state of the electronic device **101**, and may control the USB PDIC **323** to control the VBUS voltage of the PD source **350** in response to a predetermined state.
(58) According to an example embodiment, blocks included in the hardware **320** may be mapped to blocks included in the software **310**, respectively. The USB driver **312-1**, the charging driver **312-2**, and/or the PDIC driver **312-3** may perform message transmission and/or communication with one another using notify messages or a callback function.
(59) According to an example embodiment, the USB connector **330** may be connected to an external device, for example, the PD source **350**, which in turn is also capable of USB connection. The USB connector **330** may be a USB port. The USB port may be a USB Type-C port. According to an example embodiment, the electronic device **101** may include an opening formed on one surface of its housing and a hole connected to the opening, and the USB connector **330** may be disposed in the hole. A connector of the PD source **350** may be inserted into the USB connector **330** of the electronic device **101**, and may be electrically connected to the USB connector **330** due to the physical contact. According to an example embodiment, the USB connector **330** and the hole of the electronic device **101** may have a reversible structure.
(60) According to an example embodiment, one surface of the connector of the PD source **350** may be inserted into the USB connector **330** of the electronic device **101** in a direction parallel to the front surface (e.g., the surface on which a display is located) of the electronic device **101**. Similarly, another surface of the connector of the PD source **350** may be inserted in a direction parallel to the front surface of the electronic device **101**.
(61) According to an example embodiment, the PD source **350** may include a USB connector (not shown) and a USB PDIC (not shown), and may directly supply direct current (DC) power to the electronic device **101**. The PD source **350** may be a device configured to supply power to each component of the electronic device **101** through a USB Type-C PD adapter, a USB Type-C multi-adapter, or an external electronic device (e.g., the electronic device **102** of FIG. **1**). Certain example embodiments in which power is supplied from the PD source **350** to the electronic device **101** via the USB Type-C port will be described with reference to FIGS. **5**A and **5**B below.
(62) According to an example embodiment, the USB connector **330** may be set at the lowest communication speed supported by the USB standard. The lowest communication speed may be about 1.5 Mbps, full speed may be about 12 Mbps, high speed may be about 480 Mbps, and/or super speed may be 5 Gbps or more. USB 2.0 may support up to the high speed. USB 3.0 or higher may support up to the super speed.
(63) According to an example embodiment, when the USB PDIC **323** determines that the CC pin and the PD source **350** are connected, the processor **120** may control the VBUS voltage of the PD

source **350** to be greater than or equal to a threshold voltage, for example, 5 V, through the CC pin, so that power may be supplied from the PD source **350** to the electronic device **101**.
(64) According to an example embodiment, when the processor **120** receives power supplied from the PD source **350**, the USB PDIC **323** may control the VBUS voltage of the PD source **350** to be a voltage (e.g., 0 V) less than the threshold voltage (e.g., 5 V) while maintaining communication with the PD source **350** through the CC pin based on the state of the electronic device **101**.
(65) According to an example embodiment, when the electronic device **101** and the PD source **350** are connected through the USB Type-C port, the processor **120** may receive, from the PD source **350**, information about whether the VBUS voltage is controllable to be less than the threshold voltage while maintaining communication between the electronic device **101** and the PD source **350** through the CC pin.
(66) Specifically, the information about whether the VBUS voltage is controllable may be implemented as a bit added to a PD message transmitted from the PD source **350** to the electronic device **101** through the CC pin. According to an example embodiment, the PD message may include a preamble, a start of packet (SOP), a message header, a data object, a cyclic redundancy check (CRC), and an end of packet (EOP) in sequence. According to an example embodiment, the PD source **350** connected to the electronic device **101** may add one bit to the PD message to transmit the information about whether the VBUS voltage is controllable to be less than the threshold voltage while maintaining the communication through the CC pin to the electronic device **101**. For example, if the VBUS voltage is controllable to be less than the threshold voltage while maintaining the communication through the CC pin with the PD source **350**, the bit value may be set to “1” and the bit may be transmitted to the electronic device **101**. Conversely, if the VBUS voltage is not controllable, the bit value may be set to “0” and the bit may be transmitted to the electronic device **101**.
(67) According to an example embodiment, if the USB PDIC **323** determines that the VBUS voltage is controllable to be less than the threshold voltage while maintaining the communication through the CC pin, the processor **120** may control the USB PDIC **323** to control the VBUS voltage of the PD source **350** to be a voltage (e.g., 0 V) less than the threshold voltage (e.g., 5 V) while maintaining the communication through the CC pin with the PD source **350** based on the state of the electronic device **101**.
(68) According to an example embodiment, when the electronic device **101** and the PD source **350** are connected via the USB port, the processor **120** may receive the information about whether the VBUS voltage is controllable to be less than the threshold voltage while maintaining the communication between the electronic device **101** and the PD source **350** through the CC pin. According to an example embodiment, to control the VBUS voltage to be less than the threshold voltage based on the state of the electronic device **101**, the processor **120** may send a request for the information about whether the VBUS voltage is controllable to the PD source **350** and receive the information.
(69) An operation in which the processor **120** controls the VBUS voltage of the PD source **350** based on the state of the electronic device **101** according to certain example embodiments will be described in detail with reference to FIG. **7** below.
(70) FIG. **4** is a diagram illustrating pins constituting a USB port according to an example embodiment.
(71) FIG. **4** illustrates pins **411**, **412**, **421**, **422**, **423**, **424**, **431**, **432**, **433**, and **434** constituting a USB port **400** (e.g., the USB connector **330** of FIG. **3**) according to an example embodiment. The USB port according to an example embodiment may be a USB Type-C port.
(72) According to an example embodiment, the USB Type-C port may include 24 pins. The USB Type-C port (e.g., the USB connector **330** of FIG. **3**) included in an electronic device (e.g., the electronic device **101** of FIG. **3**) may include a receptacle structure having pins into which a terminal may be inserted. The USB Type-C port included in the PD source (e.g., the PD source **350**

of FIG. **3**) may include a plug structure having pins capable of being inserted into a terminal (e.g. the aforementioned receptacle). The USB Type-C port may include first pins **411** and **412**, second pins **421**, **422**, **423**, and **424**, and/or third pins **431**, **432**, **433**, and **434**. According to an example embodiment, the first pins **411** and **412** may be CC pins.
(73) The first pins **411** and **412** may include the CC1 pin **411** and the CC2 pin **412**. The CC1 pin **411** and the CC2 pin **412** may detect the mounting state of the PD source **350**, i.e., when the PD source **350** is connected using the USB Type-C port **330**. The CC1 pin **411** and CC2 pin **412** may be used for communication between the USB PDIC (e.g., the USB PDIC **323** of FIG. **3**) and the PD source **350** via the USB connector **330**. The two first pins, e.g., the first pins **411** and **412**, may be disposed to be symmetrical such that the top and the bottom of the USB cable are indistinguishable from each other, i.e. so that the USB cable is reversible.
(74) According to an example embodiment, when a resistor Rp is recognized by the CC1 pin **411** and/or the CC2 pin **412**, the electronic device **101** may operate in a USB client mode. The resistor Rp may be a resistor defined in the USB Type-C specification and may be, for example, about 10 kΩ.
(75) According to an example embodiment, the second pins **421**, **422**, **423**, and **424** may be VBUS pins. Power may be supplied from the external PD source **350** through the second pins **421**, **422**, **423**, and **424**.
(76) According to an example embodiment, each of the third pins **431**, **432**, **433**, and **434** may be a D+ pin or a D− pin. The third pins **431**, **432**, **433**, and **434** may perform USB 2.0 interface communication.
(77) According to an example embodiment, the USB Type-C port may further include pins TX1+, TX1−, TX2+, TX2−, RX1+, RX1−, RX2+, and RX2− that form a high-speed data path, auxiliary bus pins SBU1 and SBU2, and/or a ground pin GND.
(78) FIGS. **5**A and **5**B are diagrams illustrating an operation in which an electronic device **101** receives supplied power, according to certain example embodiments.
(79) FIG. **5**A illustrates an example in which the electronic device **101** and the PD source **350** outside the electronic device **101** are connected via the USB connector **330** and the electronic device **101** receives a VBUS voltage of 5 V or greater supplied from the PD source **350**.
(80) As described above with reference to FIGS. **3** and **4**, the electronic device **101** and the PD source **350** may be connected through the 24 pins of the USB connector **330**, and specifically, may perform PD-related communication through the CC pin (e.g., the CC pins **411** and **412** of FIG. **4**), and power may be supplied through the VBUS pin (e.g., the VBUS pins **421**, **422**, **423**, and **424** of FIG. **4**). Referring to FIG. **5**A, the USB connector **330** may include the VBUS pin **550** and the CC pin **560**. Power supply through the VBUS pin is indicated by a bolded line, and PD communication through the CC pin is indicated by a dashed line.
(81) According to an example embodiment, when the electronic device **101** and the PD source **350** are connected through the USB connector **330**, the USB PDIC **323** of the electronic device **101** may determine, through the CC pin **560**, whether the PD source **350** is connected through the USB connector **330**. When the USB PDIC **323** determines that the USB connector **330** is connected to the PD source **350**, the USB PDIC **323** may control the voltage of the VBUS pin **550**, which is output through an analog current (AC)-to-DC or DC-to-DC converter **510** and the USB PDIC **523** of the PD source **350**, to be greater than or equal to 5 V so that the communication through the CC pin **560** may be maintained.
(82) According to an example embodiment, when the voltage of the VBUS pin **550** supplied from the PD source **350** is greater than or equal to 5 V, the electronic device **101** may receive power of 5 V or greater supplied through the charging circuitry **210**, and convert the power using a DC-DC converter **530** so that the converted power may be supplied to various components of the electronic device **101**. FIG. **5**A illustrates power supply to the processor **120** of the electronic device **101** through the charging circuitry **210** and the DC-DC converter **530** for brevity of description,

however, this is merely an example. For example, as described above with reference to FIG. **3**, power may be supplied to each component of the electronic device **101**.
(83) According to an example embodiment, when the voltage of the VBUS pin **550** supplied from the PD source **350** is greater than or equal to 5 V, power may be supplied to each component of the electronic device **101** through the charging circuitry **210** and the battery **189** may also be charged.
(84) According to an example embodiment, the charging circuitry **210** may include the DC-DC converter **530**. For example, the charging circuitry **210** may include a buck converter, a boost converter, a buck-boost converter, and/or a Ćuk converter.
(85) According to an example embodiment, in the basic USB Type-C power delivery operation as shown in FIG. **5**A, the voltage of the VBUS pin **550** may be controlled to be in a range of 5 V to 20 V. However, conventionally since the voltage of the VBUS pin **550** needs to be maintained at 5 V or greater even though there is no need for power delivery through the VBUS, standby power may continue to be consumed.
(86) FIG. **5**B illustrates an example in which the electronic device **101** and the PD source **350** outside the electronic device **101** are connected via the USB connector **330**, and in which the VBUS voltage received from the PD source **350** by the electronic device **101** is controlled to be less than a threshold voltage (e.g., 5 V).
(87) As described above with reference to FIG. **5**A, when the electronic device **101** is connected to the PD source **350** via the USB connector **330**, the VBUS voltage may be maintained at 5V or greater. According to an example embodiment, in FIG. **5**B, when it is determined that there is no need to maintain the voltage of the VBUS pin **550** to be greater than or equal to the threshold voltage (e.g., 5 V), based on a state of the electronic device **101** such as the charge level of a battery **189** and system power state, the processor **120** may control the VBUS voltage to be a voltage (e.g., 0 V) less than the threshold voltage.
(88) According to an example embodiment, as described above with reference to FIG. **3**, when the voltage of the VBUS pin **550** is designed to be controllable to be less than the threshold voltage while the electronic device **101** maintains communication through the CC pin **560** with the PD source **350** using the USB PDIC **523**, the processor **120** may control the VBUS voltage to be less than the threshold voltage. Accordingly, the processor **120** may further perform an operation of receiving information about whether the VBUS voltage is controllable from the PD source **350**.
(89) As described above with reference to FIG. **3**, in an example, when the electronic device **101** and the PD source **350** are connected, the information about whether the VBUS voltage is controllable may be included in a PD message and transmitted from the PD source **350** to the processor **120**. In another example, when the processor **120** determines that there is no need to maintain the VBUS voltage to be greater than or equal to the threshold voltage, the information about whether the VBUS voltage is controllable may be obtained from the PD source **350** by sending a request for the information to the PD source **350**. The PD message may be transmitted through the CC pin **560**.
(90) According to an example embodiment, when the voltage of the VBUS pin **550** is controllable to be less than the threshold voltage while the electronic device **101** maintains the communication through the CC pin **560** with the PD source **350**, and when the processor **120** determines, based on the state of the electronic device **101**, that there is no need to maintain the VBUS voltage to be greater than or equal to the threshold voltage, the VBUS voltage may be controlled to be a voltage (e.g., 0 V) less than the threshold voltage (e.g., 5 V) as shown in FIG. **5**B.
(91) According to an example embodiment, when the voltage of the VBUS pin **550** supplied from the PD source **350** is 0 V, the electronic device **101** may receive power through the battery **189**, instead of the charging circuitry **210**. As described above with reference to FIG. **5**A, FIG. **5**B illustrates that power is supplied from the battery **189** to the processor **120** for brevity of description, however, this is merely an example. For example, power may be supplied to each component of the electronic device **101** from the battery **189**.

(92) Unlike the power delivery operation described with reference to FIG. 5A, in the power delivery operation of FIG. 5B, since the voltage of the VBUS pin **550** is 0 V (OFF), standby power may be reduced. Table 1 below shows standby power in an example in which the VBUS voltage is controlled to 0 V according to an example embodiment. Referring to Table 1, if there is no VBUS OFF function, standby power may range from 400 millivolts (mV) to 1 watt (W). However, it can be found that when the processor **120** controls the VBUS voltage to be less than the threshold voltage, as described above, the standby power is reduced to 65 milliwatts (mW).
(93) TABLE-US-00001 TABLE 1 Power state VBUS OFF function Standby power ON mode X 400 mV to 1 W ◯ (VBUS OFF function 65 mW active) Power-saving ◯ (VBUS OFF function 65 mW mode active with fully charged battery) Off mode ◯ (VBUS OFF function 65 mW active with fully charged battery)
(94) Table 1 shows an example in which the power state of the electronic device **101** is the ON mode and the VBUS OFF function is active, and an example in which the power state is in a power-saving mode or OFF mode, the VBUS OFF function is active, and the battery is fully charged, however, this is merely an example. Various other embodiments may be possible. The operation in which the processor **120** controls the voltage of the VBUS pin **550** of the PD source **350** based on the state of the electronic device **101** according to an example embodiment will be described in detail with reference to FIG. **7** below.
(95) <Method of Operating Electronic Device>
(96) FIG. **6** is a flowchart illustrating an operation of an electronic device according to an example embodiment.
(97) Operations **610** to **640** may be performed by the processor **120** of the electronic device **101** described above with reference to FIG. **3**, and accordingly the description provided with reference to FIGS. **1** to **5**B will not be repeated for conciseness.
(98) According to an example embodiment, in operation **610**, the processor **120** may determine whether a first pin of the USB port of the electronic device **101** is connected to the PD source **350**. As described above with reference to FIG. **3**, the processor **120** may determine whether the USB port and the PD source **350** are connected, through the USB PDIC **323**. According to an example embodiment, the USB port may be a USB Type-C port, and the first pin may be a CC pin (e.g., the CC pins **411** and **412** of FIG. **4**).
(99) According to an example embodiment, when the USB PDIC **323** determines that the first pin and the PD source **350** are connected in operation **610**, the processor **120** may determine whether the voltage of a second pin (e.g., the VBUS pin **550** of FIG. **5**B) is controllable to be less than a threshold voltage while the electronic device maintains communication through the first pin (e.g., the CC pin **560** of FIG. **5**B) with the PD source **350**, using the USB PDIC **523** in operation **620**. The second pin may be a VBUS pin (e.g., the VBUS pins **421**, **422**, **423**, and **424** of FIG. **4**). As described above with reference to FIGS. **3** and **5**B, the processor **120** may determine whether the voltage of the second pin is controllable, based on a bit included in a PD message received from the PD source **350**.
(100) According to an example embodiment, when it is determined in operation **620** that the voltage of the second pin is controllable to be less than the threshold voltage while maintaining the communication through the first pin with the PD source **350**, the processor **120** may monitor a state of the electronic device in operation **630**. Although operation **630** is performed after operation **620** as shown in FIG. **6**, the example embodiments are not limited thereto. For example, information about whether the voltage of the second pin is controllable to be less than the threshold voltage may be transmitted when the electronic device **101** and the PD source **350** are connected, as described above with reference to FIG. **3**, or when the processor **120** determines that the voltage of the second pin needs to be less than the threshold voltage while monitoring the state of the electronic device **101** and sends a request for the information.
(101) According to an example embodiment, in operation **630**, the processor **120** may monitor the

state of the electronic device **101** and determine whether to control the voltage of the second pin to be less than the threshold voltage while maintaining the communication through the first pin. According to an example embodiment, in operation **640**, the processor **120** may control the voltage of the second pin to be less than the threshold voltage while maintaining the communication through the first pin based on a monitoring result obtained in operation **630**.

(102) Certain example embodiments in which the processor **120** monitors the state of the electronic device **101** and determines whether to control the voltage of the second pin to be less than the threshold voltage while maintaining the communication through the first pin in operation **630** will be described in detail with reference to FIG. **7** below.

(103) FIG. **7** is a flowchart illustrating an operation of controlling a virtual bus (VBUS) voltage based on a state of an electronic device according to an example embodiment.

(104) Operations **710** to **730** may be performed by the processor **120** of the electronic device **101** described above with reference to FIG. **3**. According to an example embodiment, operations **710** to **730** may correspond to the operation (e.g., operation **630** of FIG. **6**) of monitoring the state of the electronic device **101** described above with reference to FIG. **6**.

(105) According to an example embodiment, when it is determined in operation **620** that the voltage of the second pin is controllable to be less than the threshold voltage while the electronic device maintains the communication through the first pin with the PD source **350**, the processor **120** may monitor the power state of the electronic device **101** and the charge level of the battery **189** to determine whether to control the voltage of the second pin to be less than the threshold voltage while maintaining the communication through the first pin. The first pin may be a CC pin (e.g., the CC pins **411** and **412** of FIG. **4**), and the second pin may be a VBUS pin (e.g., the VBUS pins **421**, **422**, **423**, and **424** of FIG. **4**).

(106) According to an example embodiment, in operation **710**, the processor **120** may monitor the power state of the electronic device **101**. The power state of the electronic device **101** may include, for example, an ON mode, a power-saving mode (e.g. sleep mode), and an OFF mode as shown in Table 1 described above with reference to FIG. **5**B. The ON mode may refer to the mode in which the electronic device **101** operates, and the power-saving mode may be a low-power mode for saving power of the battery **189**. The OFF mode may refer to the state of being powered off. For example, the power-saving mode may be a state in which at least one of components (e.g., the input module **150**, the sound output module **155**, the display module **160**, the memory **130**, the processor **120**, the communication module **190** and/or the interface **177**) of the electronic device **101** is powered off, a sleep state, or a low-power state.

(107) According to an example embodiment, when the electronic device **101** is in the OFF mode or the power-saving mode in operation **710** and when the battery **189** is fully charged in operation **720**, the processor **120** may determine that there is no need to receive power from the PD source **350** and control the voltage of the second pin to be less than the threshold voltage while maintaining the communication through the first pin, for optimization of standby power.

(108) According to an example embodiment, when the charge level of the battery **189** is greater than or equal to a set voltage in operation **730**, even though the electronic device **101** is in the ON mode, not in the OFF mode or the power-saving mode, in operation **710**, the VBUS voltage through the PD source **350** may still be unnecessary, and the processor **120** may control the voltage of the second pin to be less than the threshold voltage while maintaining the communication through the first pin, for optimization of standby power.

(109) According to an example embodiment, the set voltage of the battery **189** may refer to a set capacity of the battery **189**. For example, operation **730** of determining whether the charge level of the battery **189** is greater than or equal to the set voltage may include determining whether the battery **189** is charged to a capacity greater than or equal to a set capacity.

(110) The VBUS voltage according to an operation of the processor **120** in the example of FIG. **7** may be summarized as shown in Table 2 below.

(111) TABLE-US-00002 TABLE 2 Power state Battery capacity VBUS ON mode Less than set Greater than or voltage equal to 5 V Greater than or 0 V equal to set voltage Power-saving 100% 0 V mode Off mode 100% 0 V

(112) According to an example embodiment, based on the determination of the processor **120** described with reference to FIG. **7**, in operation **640**, the voltage of the VBUS pin (e.g., the VBUS pin **550** of FIG. **5**B) may be controlled to be a voltage (e.g., 0 V) less than the threshold voltage (e.g., 5 V) while the communication between the electronic device **101** and the PD source **350** through the CC pin (e.g., the CC pin **560** of FIG. **5**B) is maintained. Thus, unnecessary standby power in the electronic device **101** may be reduced.

(113) According to an example embodiment, an electronic device **101** may include: a USB PDIC **323** configured to determine whether a first pin (e.g., the CC pins **411** and **412** of FIG. **4**) of a USB port of the electronic device is connected to a PD source **350**; a charging circuitry **210** configured to charge a battery **189** of the electronic device **101** with power supplied from the PD source **350** through a second pin (e.g., the VBUS pins **421**, **422**, **423**, and **424** of FIG. **4**) of the USB port; a memory **130** configured to store computer-executable instructions; and a processor **120** configured to execute the instructions by accessing the memory **130**.

(114) According to an example embodiment, when the USB PDIC **323** determines that the first pin and the PD source **350** are connected, the instructions may cause the processor **120** to control a voltage of the second pin to be less than a threshold voltage while maintaining communication between the USB PDIC **323** and the PD source **350** through the first pin based on a state of the electronic device **101**.

(115) According to an example embodiment, when the USB PDIC **323** determines that the first pin and the PD source **350** are connected, the instructions may cause the processor **120** to receive information, from the PD source **350**, about whether the voltage of the second pin is controllable to be less than the threshold voltage while maintaining the communication between the USB PDIC **323** and the PD source **350** through the first pin.

(116) According to an example embodiment, the information may be included in a bit added to a message transmitted from the PD source **350** to the electronic device **101**.

(117) According to an example embodiment, when the electronic device **101** is powered off and when the battery **189** is fully charged, the instructions may cause the processor **120** to control the voltage of the second pin to be less than the threshold voltage while maintaining the communication between the USB PDIC **323** and the PD source **350** through the first pin.

(118) According to an example embodiment, when the electronic device **101** is powered on and when a voltage of the battery **189** is greater than or equal to a set voltage, the instructions may cause the processor **120** to control the voltage of the second pin to be less than the threshold voltage while maintaining the communication between the USB PDIC **323** and the PD source **350** through the first pin.

(119) According to an example embodiment, the instructions may cause the processor **120** to control the voltage of the second pin to be 0 V while maintaining the communication between the USB PDIC **323** and the PD source **350** through the first pin based on the state of the electronic device **101**.

(120) According to an example embodiment, the threshold voltage may be 5 V.

(121) According to an example embodiment, the USB port may be a USB Type-C port, the first pin may be at least one of a CC1 pin and a CC2 pin, and the second pin may be a VBUS pin.

(122) According to an example embodiment, a method of operating an electronic device **101** may include: determining whether a first pin (e.g., the CC pins **411** and **412** of FIG. **4**) of a USB port of the electronic device **101** is connected to a PD source **350**; and when it is determined that the first pin and the PD source **350** are connected, controlling a voltage of a second pin (e.g., the VBUS pins **421**, **422**, **423**, and **424** of FIG. **4**) of the USB port to be less than a threshold voltage while maintaining communication between the electronic device **101** and the PD source **350** through the

first pin based on a state of the electronic device **101**, wherein a battery **189** of the electronic device **101** is configured to receive power supplied from the PD source **350** through the second pin.

(123) According to an example embodiment, the method may further include, when it is determined that the first pin and the PD source **350** are connected, receiving information, from the PD source **350**, about whether the voltage of the second pin is controllable to be less than the threshold voltage (e.g., 5 V) while maintaining the communication between the electronic device **101** and the PD source **350** through the first pin.

(124) According to an example embodiment, the information may be included in a bit added to a message transmitted from the PD source **350** to the electronic device **101**.

(125) According to an example embodiment, the voltage of the second pin may be controlled to be less than the threshold voltage when the electronic device **101** is powered off and when the battery **189** is fully charged.

(126) According to an example embodiment, the voltage of the second pin may be controlled to be less than the threshold voltage when the electronic device **101** is powered on and when a voltage of the battery **189** is greater than or equal to a set voltage.

(127) According to an example embodiment, the voltage of the second pin may be controlled to be 0 V.

(128) According to an example embodiment, the threshold voltage may be 5 V.

(129) According to an example embodiment, the USB port may be a USB Type-C port, the first pin may be at least one of a CC1 pin and a CC2 pin, and the second pin may be a VBUS pin.

(130) According to an example embodiment, a non-transitory computer-readable storage medium may store instructions that, when executed by an electronic device, cause the electronic device **101** to: determine whether a first pin (e.g., the CC pins **411** and **412** of FIG. **4**) of a USB port of the electronic device **101** is connected to a PD source **350**; and when it is determined that the first pin and the PD source **350** are connected, control a voltage of a second pin (e.g., the VBUS pins **421**, **422**, **423**, and **424** of FIG. **4**) of the USB port to be less than a threshold voltage while maintaining communication between the electronic device **101** and the PD source **350** through the first pin based on a state of the electronic device **101**, wherein a battery **189** of the electronic device **101** is configured to receive power supplied from the PD source **350** through the second pin.

(131) According to an example embodiment, the instructions may further cause the electronic device **101** to, when it is determined that the first pin and the PD source **350** are connected, receive information, from the PD source **350**, about whether the voltage of the second pin is controllable to be less than the threshold voltage while maintaining the communication between the electronic device **101** and the PD source **350** through the first pin.

(132) According to an example embodiment, the instructions may cause the electronic device **101** to, when the electronic device **101** is powered off and when the battery **189** is fully charged, control the voltage of the second pin to be less than the threshold voltage while maintaining the communication between the electronic device **101** and the PD source **350** through the first pin.

(133) According to an example embodiment, the USB port may be a USB Type-C port, the first pin may be at least one of a CC1 pin and a CC2 pin, and the second pin may be a VBUS pin.

(134) Certain of the above-described embodiments of the present disclosure can be implemented in hardware, firmware or via the execution of software or computer code that can be stored in a recording medium such as a CD ROM, a Digital Versatile Disc (DVD), a magnetic tape, a RAM, a floppy disk, a hard disk, or a magneto-optical disk or computer code downloaded over a network originally stored on a remote recording medium or a non-transitory machine readable medium and to be stored on a local recording medium, so that the methods described herein can be rendered via such software that is stored on the recording medium using a general purpose computer, or a special processor or in programmable or dedicated hardware, such as an ASIC or FPGA. As would be understood in the art, the computer, the processor, microprocessor controller or the programmable hardware include memory components, e.g., RAM, ROM, Flash, etc. that may store

or receive software or computer code that when accessed and executed by the computer, processor or hardware implement the processing methods described herein.

(135) While the present disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the present disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a universal serial bus (USB) power delivery integrated circuit (PDIC) configured to perform communication with a power delivery (PD) source through a first pin of a USB port of the electronic device; a charging circuitry configured to charge a battery of the electronic device with power supplied from the PD source through a second pin of the USB port; memory storing computer-executable instructions; and at least one processor including processing circuitry, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: monitor a power state of the electronic device and a charge level of the battery in a state in which a voltage of the second pin is greater than or equal to a threshold voltage, and in response to the electronic device being powered off or being in a power-saving mode and the battery being fully charged, control the voltage of the second pin to be less than the threshold voltage while maintaining the communication between the USB PDIC and the PD source through the first pin.
2. The electronic device of claim 1, wherein, when the USB PDIC determines that the first pin and the PD source are connected, the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to receive information, from the PD source, about whether the voltage of the second pin is controllable to be less than the threshold voltage while maintaining the communication between the USB PDIC and the PD source through the first pin.
3. The electronic device of claim 2, wherein the information is included in a bit added to a message transmitted from the PD source to the electronic device.
4. The electronic device of claim 1, wherein the controlling of the voltage of the second pin comprises when the electronic device is powered on and when a voltage of the battery is greater than or equal to a set voltage, controlling the voltage of the second pin to be less than the threshold voltage while maintaining the communication between the USB PDIC and the PD source through the first pin.
5. The electronic device of claim 1, wherein the controlling of the voltage of the second pin comprises controlling the voltage of the second pin to be 0 volts (V) while maintaining the communication between the USB PDIC and the PD source through the first pin based on the state of the electronic device.
6. The electronic device of claim 1, wherein the threshold voltage is 5 V.
7. The electronic device of claim 1, wherein: the USB port is a USB Type-C port, the first pin includes a configuration channel (CC) 1 pin and/or a CC2 pin, and the second pin is VBUS pin.
8. A method of operating an electronic device, the method comprising: receiving a power from a power delivery (PD) source through a second pin of a USB port of the electronic device; monitoring a power state of the electronic device and a charge level of a battery of the electronic device in a state in which a voltage of the second pin is greater than or equal to a threshold voltage; and in response to the electronic device being powered off or being in a power-saving mode and the battery being fully charged, controlling the voltage of the second pin to be less than the threshold voltage while maintaining communication between the electronic device and the PD source through a first pin of the USB port.
9. The method of claim 8, further comprising: when it is determined that the first pin and the PD

source are connected, receiving information, from the PD source, about whether the voltage of the second pin is controllable to be less than the threshold voltage while maintaining the communication between the electronic device and the PD source through the first pin.

10. The method of claim 9, wherein the information is included in a bit added to a message transmitted from the PD source to the electronic device.

11. The method of claim 8, wherein the controlling of the voltage of the second pin comprises controlling the voltage of the second pin to be less than the threshold voltage when the electronic device is powered on and when a voltage of the battery is greater than or equal to a set voltage.

12. The method of claim 8, wherein the controlling of the voltage of the second pin comprises controlling the voltage of the second pin to be 0 volts (V).

13. The method of claim 8, wherein the threshold voltage is 5 V.

14. The method of claim 8, wherein: the USB port is a USB Type-C port, the first pin includes a configuration channel (CC) 1 pin and/or a CC2 pin, and the second pin is VBUS pin.

15. A non-transitory computer-readable storage medium storing instructions that, when executed by an electronic device, cause the electronic device to: receive a power from a power delivery (PD) source through a second pin of a USB port of the electronic device; monitor a power state of the electronic device and a charge level of a battery of the electronic device in a state in which a voltage of the second pin is greater than or equal to a threshold voltage; and in response to the electronic device being powered off or being in a power-saving mode and the battery being fully charged, control the voltage of the second pin to be less than a threshold voltage while maintaining communication between the electronic device and the PD source through a first pin of the USB port.

16. The non-transitory computer-readable storage medium of claim 15, wherein the instructions further cause the electronic device to, when it is determined that the first pin and the PD source are connected, receive information, from the PD source, about whether the voltage of the second pin is controllable to be less than the threshold voltage while maintaining the communication between the electronic device and the PD source through the first pin.

17. The non-transitory computer-readable storage medium of claim 15, wherein: the USB port is a USB Type-C port, the first pin includes a configuration channel (CC) 1 pin and/or a CC2 pin, and the second pin is VBUS pin.